
Web and Multimedia Technologies

Sample exam paper 1

Notes:

- The exam paper is printed double-sided on two A3 sheets
- The exam duration is 2 hours and 45 minutes
- The maximum grade achievable in the written test is 30

Università di Pavia
Master's Degree in *Computer Engineering*

Web and Multimedia Technologies exam (Professor Marco Porta)

Date: *Month day, year*

Surname _____ Name _____

Student ID _____

Always justify your answers. Fold this sheet in half with this side as the cover. If you require additional space, use the draft (which must be submitted) and any extra sheets (available on request). Ensure that your details are written on every sheet.

1. Say whether the following statements are true (T) or false (F), clearly justifying your answers (3.5)

a. *Escape sequences* are used to include "special characters" in web pages

☐ T ☐ F

b. Color coding in TV broadcast standards (PAL, NTSC, SECAM) is indirectly based on the *RGB model*

☐ T ☐ F

c. *HTML5* is a direct evolution of *XHTML 1.0*

☐ T ☐ F

d. The *spatial resolution* of an image is given by its width and height

☐ T ☐ F

e. The concept of *tweening* is specific to vector graphics only

☐ T ☐ F

f. The same *MIDI* file played on different computers may result in different sound qualities, depending on the device used

☐ T ☐ F

g. *CGI programs* and *Application Servers* are mainly used for the same purposes

☐ T ☐ F

2. With reference to *bitmap graphics*, illustrate the concepts of *antialiasing*, *dithering*, and *safe palette* (2)

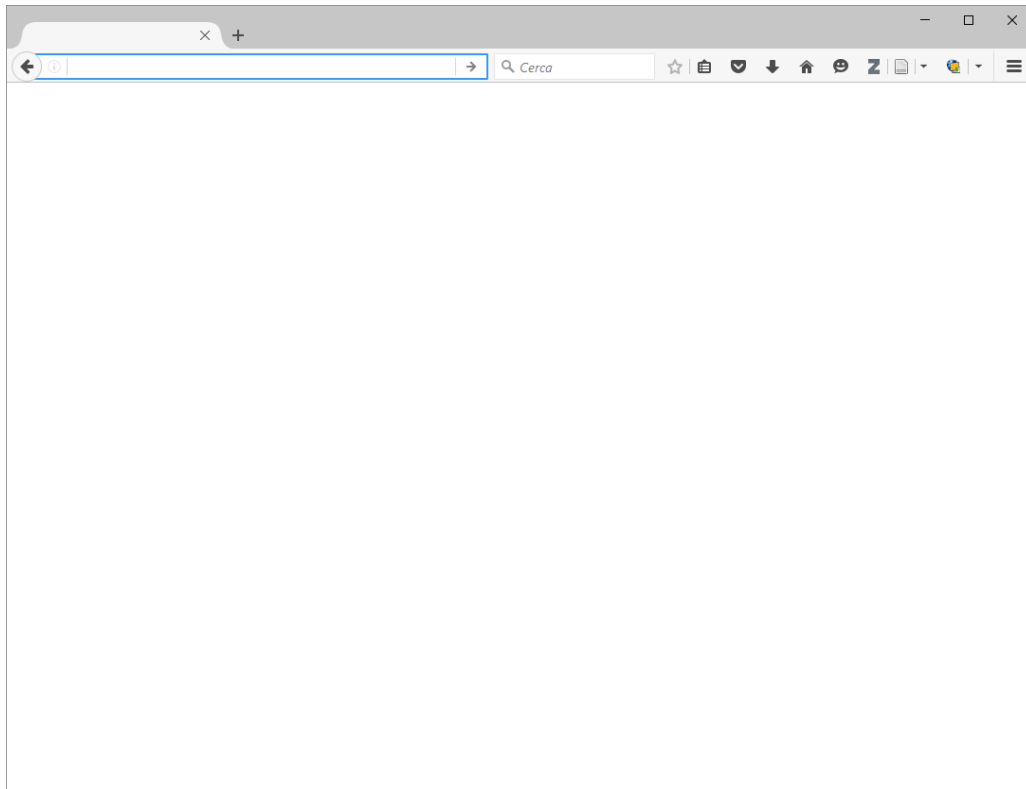
3 Given the following HTML/CSS code, provide a qualitative graphic representation of the resulting web page (5)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>University of Pavia</title>
    <style>
      body, h1 { margin: 0px; padding: 0px}
      #h {
        background-color: lightgreen; height: 70px;
      }
      #content {
        margin-left: 200px;
      }
      #menu {
        width: 190px; background-color: yellow;
        position: absolute; top: 70px; left: 0px;
      }
      td {padding: 15px}
    </style>
  </head>
  <body>
    <header id="h"><h1>Computer Engineering</h1></header>
    <article id="content">
      <table style="border: solid #999999 1px; width: 50%; margin-left: auto; margin-right: auto">
        <tr>
          <td colspan="2">
            
            The University of Pavia is one of the oldest in Europe, founded in 825.
          </td>
        </tr>
        <tr>
          <td>School of Engineering</td>
          <td>
            MSc Programs (Lauree Magistrali) in Electronic Engineering, Computer Engineering, and
            Industrial Automation Engineering
          </td>
        </tr>
      </table>
    </article>
    <div id="menu">
      <ul>
        <li><a href="syllabi/ee.html">Electronic Engineering</a></li>
        <li><a href="syllabi/ce.html">Computer Engineering</a></li>
        <li><a href="syllabi/iae.html">Industrial Automation Engineering</a></li>
      </ul>
    </div>
  </body>
</html>
```

Also, answer the following questions:

- With respect to the directory containing the HTML file defined above, where is the file `iae.html`?
- Supposing that the administrator of the web server has set `c:/internet/web/` as the root directory, what is the path (i.e., the actual position) of the file `unipv.gif` on the server? (`c:/...`)
- How would you define a form named `myform`, which, using the `get` method, sends its data to the `process.py` CGI program (with the `/cgi-bin/` absolute path)?

a.	
b.	
c.	



4. Define an external style sheet named "mystyle.css" which specifies that: (1) the page has a 30-pixel margin on all four sides and the size of `<h1>` elements is 3 times the size of the default fonts of their containers; (2) the class `gr` for the `<td>` element is characterized by a green background. Indicate (clearly): (a) the content of the CSS; and (b) the HTML code necessary to [1] load the external CSS file and to [2] make a `<td>` tag apply the `gr` class (2)

CSS content	HTML code
	1
	2

5. Showing the operations (it is not necessary to perform the calculations), indicate: (1)

a.	The size of the file resulting from the (uncompressed) digitization of 3 seconds of audio signal (mono, 12 kHz, 16 bits)
b.	The width and height (in inches) of the photograph obtained when printing an image 350 pixels wide and 200 pixels high, with a 200-dpi resolution
c.	The size of the file resulting from the (uncompressed) digitization of 1 minute of video signal (320x240 pixels, 24 bits, 15 fps)

6. With reference to an inkjet printer, what is the difference between *print* and *printer* resolutions? Why is augmenting the former too much (while keeping the latter constant) counterproductive? (2)

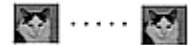
7. With the main purpose of minimizing the size of the resulting file, say whether, to implement the following animations, it is better to use a bitmap or a vector approach, or whether there is no real difference between them (justify your answers) (2)

Photograph of a cat that moves from left to right (one time)

☐ Bit.

☐ Vect.

☐ No diff.

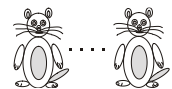


Drawing of a cat that moves from left to right (forever)

☐ Bit.

☐ Vect.

☐ No diff.



Sequence of 10 photographs (same size) appearing in the same position at regular intervals

☐ Bit.

☐ Vect.

☐ No diff.

8. (a) What is the difference between *intraframe* and *interframe* compression in digital video?
(b) Which kind of compression is used in the *DV* format, and what is its working principle?
(c) What "philosophy" is at the basis of recent developments of the MPEG format (*MPEG-7* and *MPEG-21*)? (2)

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Fold this sheet in half with this side facing up, and insert it into Sheet 1 before submission.

9. Given the following HTML/JavaScript code, concisely (but clearly) describe the meaning of each line marked with a number. Then, also explicate the "general behavior" of the page (3)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>A page containing JavaScript code</title>
    <style>
      body {background: yellow}
    </style>
    <script>
      col = new Array(3);
      col[0] = "red"; col[1] = "yellow"; col[2] = "white";
      function cs(i) {
1.      document.body.style.background = col[i];
        if (i != 2)
          document.f.display.value = col[i];
        else
2.      document.f.display.value = "";
      }
    </script>
  </head>
  <body onload="alert('Welcome! Here you can test JavaScript!')">
    <form name="f" id="f">
      <label for="display">Color</label>: <input type="text" id="display" name="display" readonly size="20">
    </form>
    <table width="300" height="100">
      <tr>
3.      <td style="background-color: red" onmouseover="cs(0)">&nbsp;</td>
        <td style="background-color: yellow" onclick="cs(1)">&nbsp;</td>
      </tr>
    </table><br />
4.    <a href="javascript:cs(2)">Reset</a><br>
  </body>
</html>
```

1.	
2.	
3.	

4.

General page behavior

10. With reference to the form defined at the previous point 9, add a text field that, when its content changes (and you click "elsewhere"), the function `check()` is executed. This function checks the content of the field and: (a) if it is 0, it substitutes it with the string `null`; (b) if it is 1, the background color of the page is changed to `cyan`; (c) if it is 2, the page text color is changed to `red`; (d) if the content of the field is neither 0 nor 1 nor 2, a message box with the warning "Wrong value!" is displayed (2.5)

11. What is a *dynamic* website? Provide at least one example of a dynamic website. What technologies are necessary to implement dynamic websites? What are their features? (2)

12. The entity *Degree course* is informally described as follows:

A degree course is characterized by a type (bachelor's or master's), a name, and a list of courses. Each course is characterized by a name, a year (1st, 2nd, etc.), a semester (1st or 2nd), and a lecturer. A lecturer may have one or more assistants.

To formally describe the entity using an XML language, define proper tags and attributes, and then use them to represent the *bachelor's degree course in "Mathematical-Literary Sciences"*, which includes, among others, the following courses:

- "Fundamentals of algebraic literature" (1st year, 2nd semester, lecturer "John Red", assistant "Arthur White")
- "Algorithmic linguistics" (2nd year, 1st semester, lecturer "Joseph Green", assistants "George Yellow" and "Mary Brown")
- "Romanticism and differential equations" (3rd year, 2nd semester, lecturer "Daniel Black", no assistants)

Apply the tags and attributes directly to the example, and then define the DTD.

Important: use at least one attribute (3)

